

SHETH L.U.J. AND SIR M.V. COLLEGE

Aim: Basic MANET Implementation with Packet Animation

Simulate a mobile ad-hoc network and visualize packet movement to study dynamic topology formation.

Manet.tcl

```
# =====  
# Basic MANET Simulation with Packet Animation  
# =====  
  
set val(chan)    Channel/WirelessChannel  
set val(prop)    Propagation/TwoRayGround  
set val(netif)   Phy/WirelessPhy  
set val(mac)     Mac/802_11  
set val(ifq)     Queue/DropTail/PriQueue  
set val(ll)      LL  
set val(ant)     Antenna/OmniAntenna  
set val(ifqlen)  50  
set val(nn)      5  
set val(rp)      AODV  
set val(x)       500  
set val(y)       500  
set val(stop)    30  
  
# Simulator  
set ns [new Simulator]  
  
# Trace Files  
set tracefd [open manet.tr w]  
$ns trace-all $tracefd  
  
set namfd [open manet.nam w]  
$ns namtrace-all-wireless $namfd $val(x) $val(y)  
  
# Topography  
set topo [new Topography]  
$topo load_flatgrid $val(x) $val(y)  
  
# GOD Object  
create-god $val(nn)  
  
# Channel  
set chan_1_ [new $val(chan)]  
  
# Wireless Configuration  
$ns node-config \  
-adhocRouting $val(rp) \  
-llType $val(ll) \  
-macType $val(mac) \  
-ifqType $val(ifq) \  
-ifqLen $val(ifqlen) \  

```

YUKTA SONAWANE T122

WSN PRACTICAL NO 15

SHETH L.U.J. AND SIR M.V. COLLEGE

```
-antType $val(ant) \  
-propType $val(prop) \  
-phyType $val(netif) \  
-channel $chan_1_ \  
-topoInstance $topo \  
-agentTrace ON \  
-routerTrace ON \  
-macTrace ON \  
-movementTrace ON
```

```
# Create Nodes  
for {set i 0} {$i < $val(nn)} {incr i} {  
    set node_i [ns node]  
}
```

```
# Initial Positions  
$node_(0) set X_ 50  
$node_(0) set Y_ 50  
$node_(0) set Z_ 0
```

```
$node_(1) set X_ 150  
$node_(1) set Y_ 150  
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 250  
$node_(2) set Y_ 250  
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 350  
$node_(3) set Y_ 350  
$node_(3) set Z_ 0
```

```
$node_(4) set X_ 450  
$node_(4) set Y_ 450  
$node_(4) set Z_ 0
```

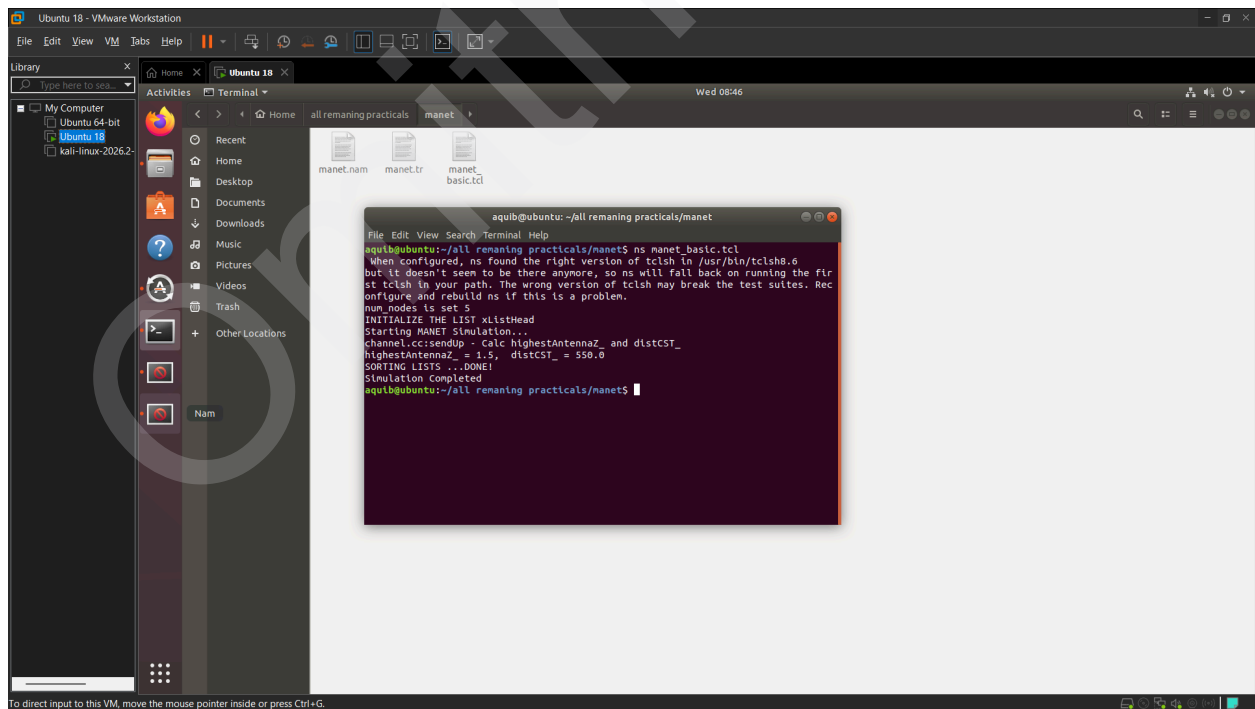
```
# Display Nodes in NAM  
for {set i 0} {$i < $val(nn)} {incr i} {  
    node i) random-motion 0  
    $ns initial_node_pos node i) 20  
}
```

```
# Mobility  
nsat2.0"node_(0) setdest 400 100 15"  
nsat4.0"node_(1) setdest 450 250 10"  
nsat6.0"node_(2) setdest 100 400 12"
```

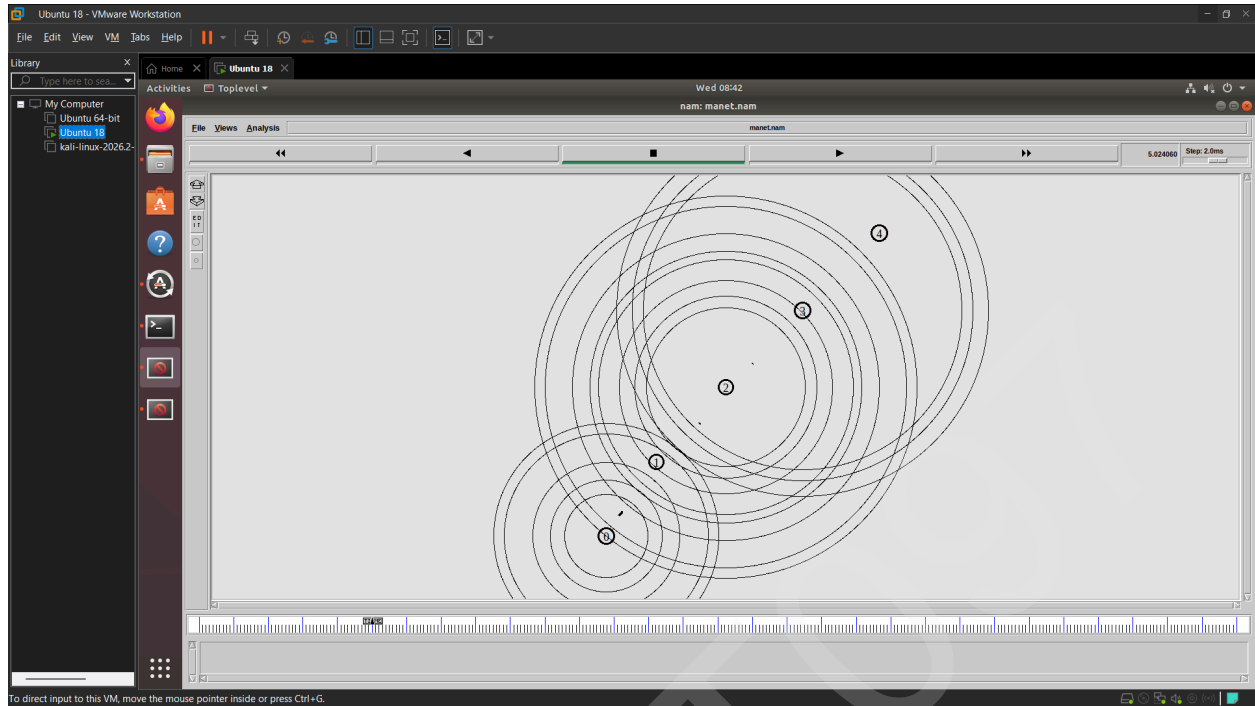
```
# UDP Agent  
set udp0 [new Agent/UDP]  
$ns attach-agent $node_(0) $udp0
```

SHETH L.U.J. AND SIR M.V. COLLEGE

```
set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0
$ns connect $udp0 $null0
# CBR Traffic
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set rate_ 200k
$cbr0 attach-agent $udp0
nsat5.0 "cbr0 start"
nsat25.0 "cbr0 stop"
# End NAM
$ns at $val(stop) "ns nam-end-wireless $val(stop)"
# Finish Procedure
proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    puts "Simulation Completed"
    exec nam manet.nam &
    exit 0
}
$ns at $val(stop) "finish"
puts "Starting MANET Simulation..."
$ns run
Output:
```



SHETH L.U.J. AND SIR M.V. COLLEGE



Result The MANET network was successfully simulated using NS2. Dynamic node movement and packet transmission were observed through NAM animation. The AODV routing protocol successfully established routes between mobile nodes.